

Predicting Customer Churn

An E-Commerce SQL Analysis — From Raw CSV to Actionable Insight

This project explores a 6,000-customer e-commerce dataset using SQL Server to identify what actually predicts customer churn. Starting from raw data with no prior SQL experience, the analysis progresses from basic filtering to multi-table joins, aggregate comparisons, and custom risk segmentation, arriving at a clear, decision-ready finding.

Tools & Skills Demonstrated

- SELECT, WHERE, ORDER BY — core filtering and sorting
- JOIN ... ON — combining two related tables via a shared key
- GROUP BY (single and multi-column) with COUNT() and AVG()
- Window functions — OVER (PARTITION BY ...) for in-group percentages
- CASE WHEN — deriving custom categorical segments from raw values
- CTEs (WITH) and HAVING — structuring multi-step analytical queries

Dataset

Two related tables, joined on a shared **Customer_ID**: **ecommerce_customer_features** (6,000 rows — behavioral and transactional data: order history, returns, engagement, satisfaction) and **ecommerce_customer_targets** (6,000 rows — binary churn outcome).

1. Establishing the Baseline

Before investigating causes, the first step was measuring the scale of the problem: what share of the customer base has actually churned?

```
SELECT churned, COUNT(*)
FROM customer_targets
GROUP BY churned;
```

Churned	Customers
No	5,071
Yes	929

Finding: 15.5% of customers (929 of 6,000) have churned. This baseline rate is the benchmark against which every subsequent factor is compared — any subgroup churning well above or below 15.5% signals a meaningful driver.

2. Comparing Behavior: Churned vs. Retained

Joining the two tables made it possible to compare average behavioral metrics between churned and retained customers side by side.

```
SELECT
    t.churned,
    COUNT(*) AS num_customers,
    AVG(f.return_rate) AS avg_return_rate,
    AVG(f.satisfaction_score) AS avg_satisfaction
FROM dbo.ecommerce_customer_features f
JOIN dbo.ecommerce_customer_targets t
    ON f.Customer_ID = t.Customer_ID
GROUP BY t.churned;
```

Metric	Retained	Churned	Gap
Return rate	6.9%	8.4%	Small
Satisfaction score	8.14	7.68	Moderate
Engagement score	5.35	2.36	Large
Cart abandonment rate	0.60	0.60	None
Days since last purchase	20 days	81 days	Very Large
Browsing frequency / week	3.17	2.56	Small-Moderate

Key insight: Metrics often assumed to drive churn — return rate and cart abandonment — showed weak or no relationship to churn in this dataset. The strongest signals were behavioral disengagement: customers going quiet (low engagement, long gaps since last purchase) far more than customers having a bad transactional experience.

3. Loyalty Program Impact

Grouping by two columns at once (loyalty membership and churn status) tested whether the loyalty program itself is associated with retention.

```
SELECT
  f.loyalty_member,
  t.churned,
  COUNT(*) AS num_customers,
  ROUND(100.0 * COUNT(*) / SUM(COUNT(*))
        OVER (PARTITION BY f.loyalty_member), 1) AS pct_within_group
FROM dbo.ecommerce_customer_features f
JOIN dbo.ecommerce_customer_targets t
  ON f.Customer_ID = t.Customer_ID
GROUP BY f.loyalty_member, t.churned
ORDER BY f.loyalty_member, t.churned;
```

Loyalty Member	Churn Rate	Sample Size
No	16.9%	4,915
Yes	9.0%	1,085

Finding: Loyalty program members churn at roughly **half the rate** of non-members (9.0% vs 16.9%) — one of the clearest and most actionable levers identified in this analysis.

4. Building a Risk Segmentation Model

Since days-since-last-purchase showed the strongest single relationship to churn, it was used to build a three-tier risk segment using CASE WHEN, structured cleanly with a CTE (WITH clause).

```
WITH customer_risk AS (  
    SELECT  
        f.Customer_ID,  
        t.churned,  
        CASE  
            WHEN f.days_since_last_purchase <= 30 THEN 'Active'  
            WHEN f.days_since_last_purchase <= 60 THEN 'At Risk'  
            ELSE 'High Risk'  
        END AS risk_segment  
    FROM dbo.ecommerce_customer_features f  
    JOIN dbo.ecommerce_customer_targets t  
        ON f.Customer_ID = t.Customer_ID  
)  
SELECT  
    risk_segment,  
    COUNT(*) AS num_customers,  
    SUM(CASE WHEN churned = 1 THEN 1 ELSE 0 END) AS num_churned,  
    ROUND(100.0 * SUM(CASE WHEN churned = 1 THEN 1 ELSE 0 END)  
        / COUNT(*), 1) AS churn_rate_pct  
FROM customer_risk  
GROUP BY risk_segment  
HAVING COUNT(*) > 100  
ORDER BY churn_rate_pct DESC;
```

Risk Segment	Definition	Customers	Churned	Churn Rate
High Risk	60+ days inactive	781	688	88.2%
At Risk	31-60 days inactive	1,371	234	17.1%
Active	0-30 days inactive	3,848	7	0.2%

Headline finding: Recency of purchase is an almost direct predictor of churn in this dataset. Customers inactive for 60+ days churn at **88.2%**, compared to essentially **0.2%** for customers active within the last 30 days — a 400x difference between the two extremes.

5. Business Takeaways

- **Recency beats everything.** Days since last purchase is the single strongest churn predictor found — far more useful than transactional friction metrics like returns or cart abandonment.
- **Engagement decays before churn is visible.** Engagement score gaps (2.36 vs 5.35) suggest customers disengage well before their churn becomes official — an early warning signal, not just a symptom.
- **Loyalty programs work.** Members churn at roughly half the rate of non-members, suggesting loyalty enrollment is a meaningful retention lever worth expanding.

- **Intuitive suspects can be red herrings.** Cart abandonment showed zero relationship to churn despite seeming like an obvious risk factor — a reminder to validate assumptions against real data rather than instinct.
- **Recommended action:** Target the 'At Risk' segment (31-60 days inactive, 17.1% churn) with re-engagement campaigns before customers cross into the 'High Risk' zone, where churn becomes near-certain.

Analysis performed end-to-end in Microsoft SQL Server (SSMS), starting from raw CSV import through final risk segmentation query. Full query set available on request.